

Institute of Actuaries of India

Subject ST7 – General Insurance: Reserving & Capital Modeling

April 2016 Examination

INDICATIVE SOLUTION

Solution 1:**i) Projecting Incurred claims data only**

- Incurred projections take into account both paid data and claim estimates.
- Incurred claims development are more mature than paid claims development.
- Projecting only incurred claims data is preferable compared to only projecting paid data.
- Paid projections can help identify changes or inconsistencies in the strength of case reserves or possible redundancies.
- Need paid claims development if discounted reserves are required.
- Projecting both bases can therefore reveal features of claim reserves that would otherwise be missed.
- Would recommend that the client projects both paid and incurred data.

Projecting gross of reinsurance and apply factor

- Projections at a gross level allows to take into account changes in reinsurance structures over the years.
- Analyze the trend in reinsurance to gross ratios for premiums, paid, incurred and outstanding claims in order to select reinsurance IBNR ratios. In addition, understand change to the reinsurance program and how this will impact the IBNR ratios selected.
- Alternatively, project at a gross level and apply the actual reinsurance program to the projected future claims.
- However, this may not be feasible in practice.

ELR using 3-year rolling average

- 3-year rolling average smooths out volatility within historical selected ultimate loss ratios.

- A weighted average e.g. ultimate premium helps further smooth out any volatility over the accident / underwriting year (as the years with the largest ultimate premiums will be more stable)
- Gives credit to the account's own unique experience.
- Easy to apply
- It can be fairly mechanical approach, which doesn't take into account any possible trends showing in the data/unusual years etc.
- Claims and premiums are calculated on the same basis.
- No account has been made to allow for premium rate changes and claims inflation.
- Would recommend a weighted average where the number of prior year historical selected ultimate loss ratios is based on the data.
- Would recommend loss ratios are adjusted to allow for premium rate changes and claims inflation.
- Recommend use of market stats benchmarks, where available for a sense check – although need to ensure that the data is on a consistent basis.

Exchange rate conversion

- Exchange rate conversion would be relevant in respect of claims where costs are driven by imports e.g. imported motor spare parts, imported capital machinery, marine cargo, aviation etc.
- The exchange rate conversion method need to be applied to paid data only and not on incurred.
- For outstanding claims, one needs to consider whether they are:
 - New claims in the quarter/year
 - Prior claims with no movement in the quarter/year
 - Re-stated claims with movement in the quarter (e.g. due to additional information or currency depreciation)

- The existing method can be applied for the first two type of outstanding claims, for the last type of claim the entire outstanding amount would need to be converted into US\$'s.
- The method assumes that the movement in exchange rates completely offsets claims inflation due to depreciation – this is not always the case in practice.
- Dealing with exchange rates is always complicated. This approach does have its disadvantages but it is not unreasonable therefore no recommendation to change the current method.

[11]

ii)

- Method can be used to reflect underlying rate changes / inflation effects and claim trends
- The method takes into account underwriter's knowledge of the business i.e. expert opinion
- Suitable for a new book of business with no prior data or extremely volatile own history – as is the case for this company.
- Can provide independent estimates
- Underwriter's estimates may be too optimistic and hence not representative of the actual loss ratios.
- Need to check consistency of the basis used with overall reserving methodology, gross/net commission/reinsurance.
- This approach is reasonable and would recommend the use of market statistics / data as well, where available, as a sense check

[3]

iii) **Motor OD**

- A short-tailed class.
- The age development factor is 80% at the end of one year which suggests that the Chain Ladder estimate has reasonable credibility.

- Also, the incurred loss ratio at the end of one year is already greater than the ELR of 70% suggesting that the ELR may not be a reasonable estimate.
- Hence the Chain Ladder estimates are more appropriate than BF ultimates.
- The incurred loss ratio has significantly surpassed the ELR suggesting that the class could have plausibly affected by CAT event e.g. Flood.
- The appropriateness of the development pattern needs to be ascertained before using the development factor to determine ultimates.
- The incurred claims excluding cat losses can be projected and the CAT losses can be projected using a different pattern if needed. The two can be added to determine the ultimate losses.

Product Liability

- A long-tailed class.
- The age development factor is only 31% at the end of...
- one year which suggests there is low credibility on the Chain Ladder ultimate.
- The BF ultimate is therefore likely to have greater reliability than chain ladder ultimate, subject to ELR being a reasonable estimate.

Crop Insurance

- Relatively short-tailed class but catastrophic exposed (no-mark if only short tailed in mentioned)
- The development factor is less than 1.
- The BF formula should therefore not be used without careful consideration as it is effectively a credibility formula between the chain ladder ultimate figure of 1,780 and $ELR \times P$ figure 990
- The BF ultimate of 1,953 is outside the range of the two figures.

[7]

[21 Marks]

Solution 2:**i) Reinstatements and Reinstatement Premium**

- An excess of loss reinsurance cover normally allows the reinsured to make only one claim on the policy.
- However, the cover can be restored after a claim, so that the reinsured still has protection. This is called reinstatement.
- Normally there will be a limit on the number of times the cover may be reinstated.
- A reinstatement may be free, or a premium may be payable.
- If premium is payable, it will normally be a proportion of the original premium.
- It will be payable whether or not the whole of the cover has been exhausted,
- ...and will be proportional to the amount of the claim to the layer.

[3]

ii)

- Rate on Line is determined as ratio of Original premium to width of layer.
- It does not include reinstatement premium in calculation

[1]

[4 Marks]**Solution 3:**

A reinsurer would analyze the claims data for the following reasons:

- Estimate the cost of outstanding claims to set reserves
- Monitor the actual runoff of outstanding claims against estimated amounts
- Assess the accuracy of the company's case estimate and/or IBNR claims reserve estimation in previous year-end exercises
- To value an insurer/reinsurer for purchase, sale or merger
- Monitoring the adequacy of its retro cover
- To provide information to management on how areas of the business are performing, and provide an indication on the profitability of business currently being written

- Reviewing current premium rates
- Determine market trends / claims inflation / new claim types
- To improve policy wordings
- Pricing new products
- Financial planning
- Monitoring the asset-liability position

[5 Marks]

Solution 4:

i) Solvency ratio = Free Reserves/Net Written Premium

Assuming no change in NWP from 2015 so NWP = NEP and net premium/incurred is net of reinsurance not acquisition costs.

Solvency Ratio (A) = $1,200/840 = 142.857\%$

Solvency Ratio (B) = $1,800/1,450 = 124.138\%$

Gross Claims ratio = Gross claims incurred/Gross Earned Premium

Net Claims ratio = Net claims incurred/Net Earned Premium

Assume ratio expressed on premium including acquisition costs.

Gross Claims ratio (A) = $800/1200 = 66.667\%$

Gross Claims ratio (B) = $1400/1500 = 93.333\%$

Net Claims ratio (A) = $720/840 = 85.714\%$

Net Claims ratio (B) = $800/1450 = 55.172\%$

ROCE = Profit Before Tax/Free Reserves at start of year

Assume free reserves at start of year = free reserves at end of year – profit in year, assuming no dividends or tax

Where Profit = net earned premium - net incurred claims - commission - other expenses
+ investment income

$$\text{ROCE (A)} = (840 - 720 - 100 - 180 + 180) / (1200 - 20) = 1.695\%$$

$$\text{ROCE (B)} = (1,450 - 800 - 100 - 105 + 300) / (1800 - 745) = 70.616\%$$

[6]

ii) Solvency Ratio

Both companies overall have a strong solvency ratio.

However, solvency ratio is unusually higher for company B

- inefficient use of capital
- looking for acquisition / growth opportunities in the market
- to cover for the uncertainty in the outstanding claims reserves, which are relatively higher
- Due to the type of business being underwritten
- ...CAT exposed
- ...this is supported by company retaining almost 97% of written premium
- ...and does not use much of reinsurance cover in terms of premium
- ...relatively low UPR as compared to Company A

A is expanding so is using free reserves to fund growth

- ...this is supported by UPR being large compared to outstanding reserves

Possible that the valuation basis for assets and liabilities is different for A & B.

Claims Ratio

Company A has a much better Gross claims ratio than Company B.

B may be worse due to poor underwriting/inadequate pricing.

B could also have been adversely impacted by catastrophe event.

Company B has a much better Net claims ratio than Company A.

Company A might not have an efficient reinsurance structure.

Company B seems to have made a huge recovery.

Different reinsurance strategy

e.g. Company B has a much higher retention strategy as compared to Company A

Different reserving philosophy

e.g. stronger case reserves or reserve margins used in Company B

Different types of business written.

e.g. longer tail, different claims trends etc

ROCE

Company B has a much higher return on capital as compared to Company A

... This is mainly on account of huge claims recovery.

Company A's ROCE also affected by the fact that is over-capitalized

Other

Reinsurance

Company A uses more reinsurance than company B

..but gets poor value from it (RI LR 20.45% vs 1200%)

A may have purchased more reinsurance to try and mitigate potential poor performing business

May have purchased reinsurance for different purposes (cat/working layers/QS)

Evidence suggests B is using mainly non-proportional

Investments

Company B has achieved marginally higher return on investments than A

...could be due to higher exposure to slightly higher exposure to equity investment.

Relative size of company

B only slightly larger balance sheet than A but writing almost 70% more net premium.

Could indicate that A is trying to reduce business to return to profitability.

Expenses

B pays slightly lower commission than A (6.67% vs 8.33%).

May indicate different distribution mix channel e.g. Direct vs B broker.

Other expenses show B appears to be more efficient than A.

[14]

[20 Marks]

Solution 5:

i)

For a lognormal, X , with parameters (μ, σ) it is known that

$$E(X) = e^{\mu + \frac{1}{2}\sigma^2}$$

$$\text{and, } E[X^2] = e^{2\mu + 2\sigma^2}$$

$$\therefore \text{Var}(x) = e^{2\mu + \sigma^2}(e^{\sigma^2} - 1)$$

Therefore,

$$\sigma = \sqrt{\ln \left(1 + \left(\frac{SD X}{E(x)} \right)^2 \right)}$$

$$\mu = \ln(E(x)) - \frac{\sigma^2}{2}$$

For Motor

$$\sigma = \sqrt{\ln \left(1 + \left(\frac{40}{75} \right)^2 \right)} = 0.5$$

$$\mu = \ln(0.75) - \frac{0.2503}{2} = -0.413$$

For Health

$$\sigma = \sqrt{\ln \left(1 + \left(\frac{35}{60} \right)^2 \right)} = .541$$

$$\mu = \ln(0.6) - \frac{0.2929}{2} = -0.657$$

To get 99.5 percentile of loss ratio ($LR_{99.5}$), we evaluate:

$$\begin{aligned} & \text{Prob}(LR < LR_{99.5}) = .995 \\ \Rightarrow & \text{Prob}\left(\frac{\ln(LR) - \mu}{\sigma} < \frac{\ln(LR_{99.5}) - \mu}{\sigma}\right) = 0.995 \\ \Rightarrow & \Phi^{-1}(0.995) = 2.576 = \frac{\ln(LR_{99.5}) - \mu}{\sigma} \\ \Rightarrow & LR_{99.5} = e^{\mu + 2.576\sigma} \end{aligned}$$

$$\text{Motor } LR_{99.5} = e^{\mu + 2.576\sigma} = 240\%$$

$$\text{Health } LR_{99.5} = e^{\mu + 2.576\sigma} = 208\%$$

$$\text{Premium risk capital} = LR_{99.5} * GWP + \text{Operating Expense ratio} * GWP - GWP$$

$$\text{Premium Risk Capital (Motor)} = 240\% * 250 + 18\% * 250 - 250 = 395.32 \text{ cr.}$$

$$\text{Premium Risk Capital (Health)} = 208\% * 250 + 18\% * 250 - 250 = 317.32 \text{ cr}$$

Assuming that there is a linear relationship between Motor and Health Losses,

$$\begin{aligned} & \text{Capital Requirement at combined level} = \\ & \sqrt{(\text{Capital Motor})^2 + (\text{Capital Health})^2 + 2 * \text{Correlation} * (\text{Capital Motor}) * (\text{Capital Health})} \\ & = 554.2 \end{aligned}$$

[12]

ii)

Market Risk Capital

Equity: The relevant scenario of return for 99.5 % risk capital is -10%

Since beta of company's portfolio is 1.2 so the return applicable by CAPM is calculated below assuming Risk free return in 99.5 % scenario is 6%.

$$\text{Return} = 6\% + 1.2 * (-10\% - 6\%) = -13.2\%$$

$$\text{Portfolio Value} = 50 * (1 - 13.2\%) = 43.4 \text{ Cr.}$$

Bond: The relevant scenario is when the interest rate is 12%

$$\text{Bond value in that scenario} = 325 / (1 + 12\%)^9 = 117.2$$

Therefore,

$$\text{Equity Capital Requirement} = 50 - 43.4 = 6.6$$

$$\text{Bonds Capital Requirement} = 150 - 117.2 = 32.8$$

$$\text{Diversified Market Risk} = \sqrt{(6.6)^2 + (32.8)^2 + 2 * 0.6 * (6.6) * (32.8)} = 37.14$$

[7]

iii) Overall Capital Requirement

$$= \sqrt{(37.14)^2 + (554.2)^2 + 2 * 0.3 * (37.14) * (189.113)} \\ = 566.5$$

[2]

iv)

$$\text{Diversification Benefit} = \text{Total of Standalone Capital} - \text{Diversified Capital} \\ = 37.14 + 554.2 - 566.5 = 24.9$$

Diversification benefit arises because the various risks in a general insurance company's operations are not perfectly correlated.

Since risks are not perfectly correlated or 100% correlated, the sum of the standalone capital requirements for each risk will usually be greater than the capital requirement for all risks combined.

Correlation can arise between class of business, policy years etc.

Generally, the diversification benefit will arise if the correlation between different risks or portfolio is

....negatively correlated. This means that 2 risk behaves in opposite directions.

For example, in the case of economic recession inflation increases which increases the average claims cost for motor damage but on the other hand people drive less which reduces the frequency of claims.

....completely uncorrelated. This means that events in two risks are completely independent.

For example, earthquake in Gujrat and forest fire in Himachal.

....positively correlated but assuming they are not perfectly correlated or 100% correlated. This means that two risks have some common driving factors but they are completely dependent.

For example, during economic downturn we expect both inflation and theft rates to increase but the change in both of these factors may be completely dependent to each other.

Good candidate will stress on the fact that.....

Operations of a general insurance company are generally positively correlated but not perfectly correlated. Good companies try to maximize the diversification benefit by writing uncorrelated

or negatively correlated business but due to complex nature of insurance business this may not be possible and we may always see a little implicit correlation between various risks such as macroeconomic reasons.

.....Law of Large numbers. Everything does not go wrong at the same time.

[5]

v) Capital allocation method:

- Pro rata on standalone basis – Diversified capital is allocated based on the ratio of standalone capital
- Shapley approach – It's a game theory approach, Marginal capital is allocated based total of all various sequence of adding the risk type to portfolio.
- Other approaches

Allocation on Prorated basis

Risk Type	Standalone Capital	Allocated Capital
Market	37.1	35.6
Premium Risk	554.2	530.9
Overall	566.5	566.5

Shapley Method

Order	Market Risk	Premium Risk
First	37.14	554.2
Second	12.3	529.3
Total	49.4	1083.5

Ratio of Market to Premium Risk	0.0456
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Allocated capital	
Market Risk	24.7
Premium Risk	541.8

[8]

[34 Marks]

Solution 6:

i)

- The model projects asset proceeds and liability outgo simultaneously based on a deterministic or stochastic model.
- Free reserves and their proceeds are treated as a balancing item.
- Different investment mixes are run through the model to determine asset mixes which give sufficient proceeds to meet liability outgo at all times.
- The aim is to determine the optimal investment strategy in line with the insurer's investment goals.
- The model may include new business, in which case it is more complex.
- Stress and scenario testing can also be used to set or test an investment strategy.

[3]

ii) Various metrics that could be used are:

- Compare the capital requirement emerging based on alternative investment strategies and select the one with minimum capital
- Compare the expected profitability of the company based on alternative investment strategies and select the one with maximum return
- Compare the Return on (Economic) Capital of the Company (ROC) and select one with maximum ROC
- Calculate the economic value of the company under different strategy and select one with highest positive impact from base scenario. Economic value could be calculated using firm's selected hurdle rate.

[4]

[7 Marks]**Solution 7:**

i) Director's and Officer's liability

[1]

ii) a)

Personal financial liability:

- Damages
- Out-of-court settlements
- Legal defence costs

For Directors and Officers liability arises from a 'wrongful act'

- Failing to comply with duties
- Errors e.g. mis-stating financial results

- Negligence
- Breach of authority

[3]

b) General exclusions :

- Fraud
- Intentional non-compliant acts
- Illegal remuneration or personal profit
- Property damage and bodily harm (except Corporate Manslaughter)
- Legal action already taken when the policy begins
- Claims made under a previous policy
- Claims covered by other insurance
- Other relevant exclusions

[3]

c) Claims made basis, some of the reasons could be

- Helps shorten the tail of the class by limiting it to claims made in current period than occurrence
- Helps in pricing the product by limiting the uncertainty from impact of adverse judicial awards
- Lesser uncertainty from claimant's behavior
- Other relevant reasons

[2]

[9 Marks]
